

Question 1



1 mark 101

A soccer team consists of 14 players. Determine the number of ways a captain and an assistant captain can be selected.

Question 2



2 marks 102

The pH of a person's blood can be found using the formula

$$\text{pH} = 6.1 + \log\left(\frac{B}{C}\right)$$

where B represents the concentration of bicarbonate in the blood, in mEq/L

C represents the concentration of carbonic acid in the blood, in mEq/L

Kansas has a blood pH of 7.41. In her blood sample, the concentration of carbonic acid is 1.41 mEq/L. Determine the concentration of bicarbonate in her blood.

Express the answer correct to one decimal place.

Question 3

3 marks 103

Solve $(2 \tan x - 1)(\tan x + 1) = 0$ where $x \in \mathbb{R}$.

Question 4



3 marks 104

Determine and simplify the 7th term in the binomial expansion of $\left(\frac{2}{x^2} + 3x\right)^9$.

Question 5



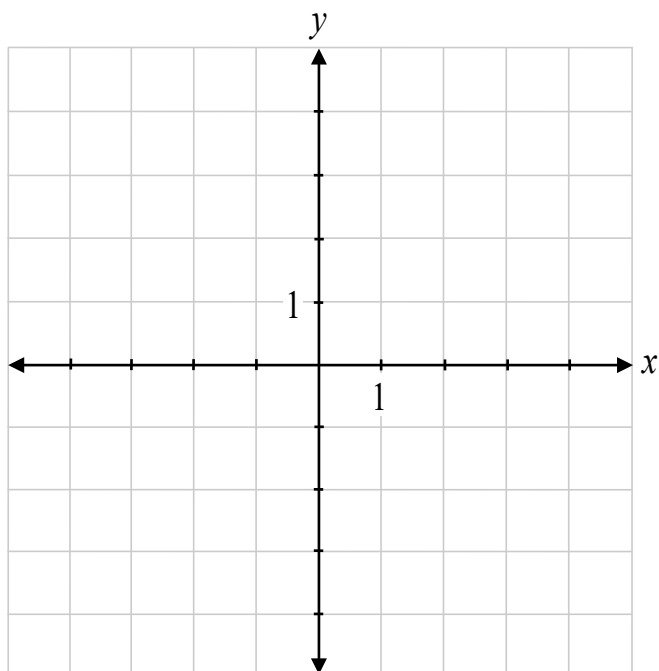
2 marks 105

A winch, with a diameter of 14 cm, is used to pull a boat out of the water. As the winch rotates, it pulls in a cable. Determine the length of cable that is pulled in if the winch rotates 526° .



Note: A calculator is not required for the remaining test questions.

Sketch the graph of $f(x) = \sqrt{\frac{1}{3}x} - 1$.



Question 7

3 marks 107

Given $y = f(x)$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection over the y -axis
- vertical stretch by a factor of 3
- horizontal translation 2 units left

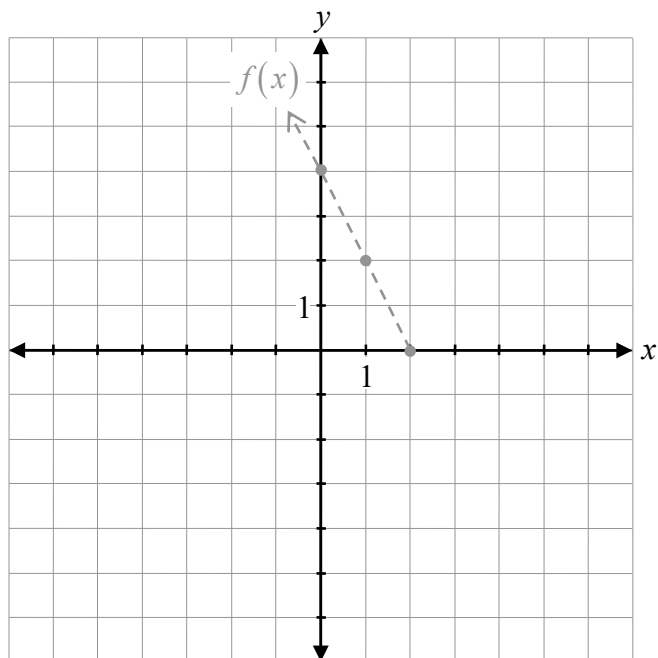
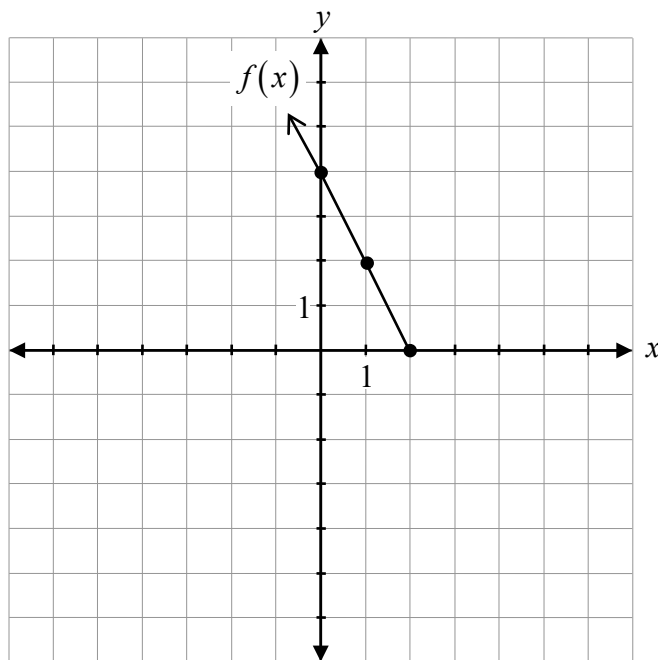
$g(x) =$ _____

Explain why the graph of $y = \frac{6x+7}{3x+5}$ has a horizontal asymptote at $y = 2$.

Question 9

3 marks 109

Given the graph of $y = f(x)$, sketch the graph of $y = -\sqrt{f(x)} + 2$.



The graph of $f(x)$ has already been drawn for your reference. No marks will be awarded for the graph of $f(x)$.

Question 10

3 marks 110

Using the laws of logarithms, completely expand the given expression.

$$\log\left(\frac{A}{\sqrt[3]{B} \cdot C^4}\right)$$

Question 11

3 marks 111

Prove the following identity for all permissible values of θ .

$$\cot \theta - \tan \theta = \frac{2 \cos 2\theta}{\sin 2\theta}$$

Left-Hand Side	Right-Hand Side

Question 12

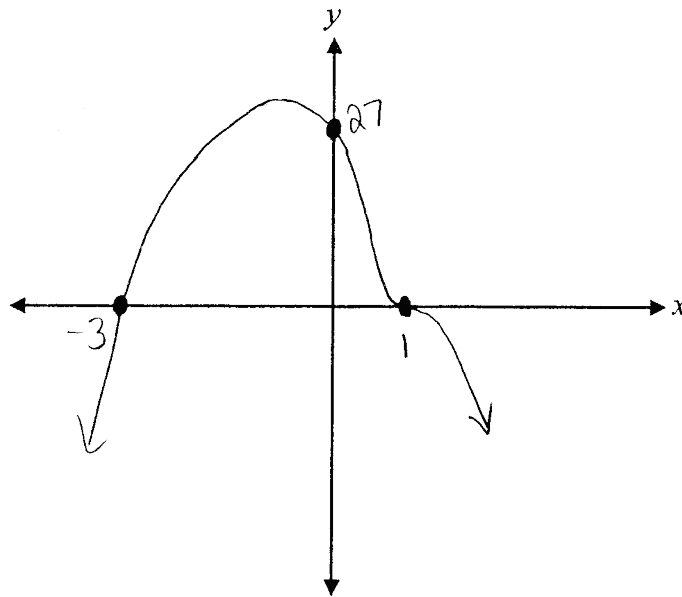
3 marks 112

Express $p(x) = -2x^3 + x^2 + 13x + 6$ in completely factored form.

$p(x) =$ _____

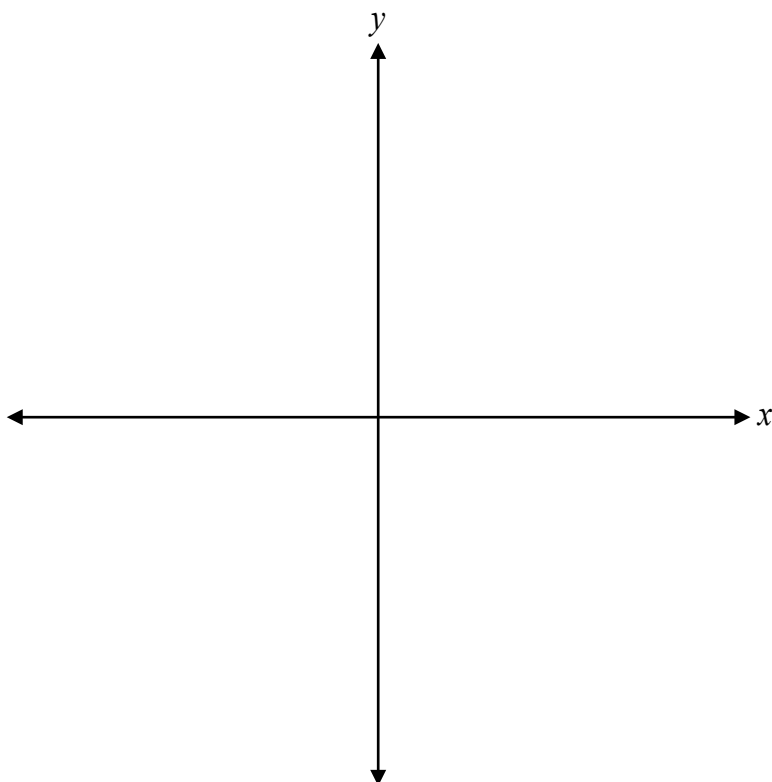
Kaitlyn was asked to sketch the graph of the polynomial function, $p(x) = -(x-1)(x+3)^3$.

Kaitlyn's graph:



Describe an error on her graph.

Sketch the angle of -5 radians in standard position.

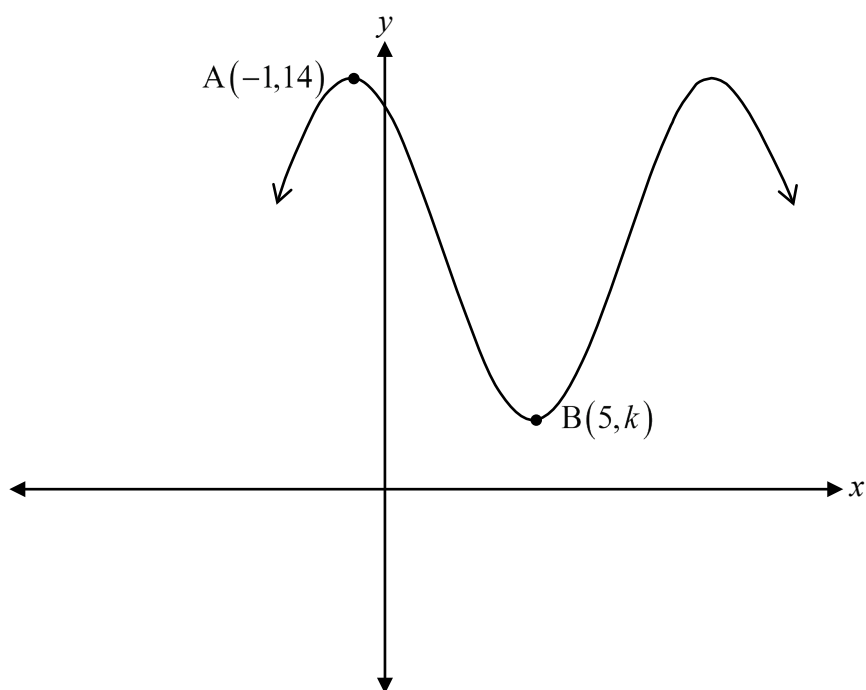


Question 15

1 mark

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The graph of a sinusoidal function is sketched below. The point A is a maximum and the point B is a minimum. State the value of k if the amplitude of the function is 6.

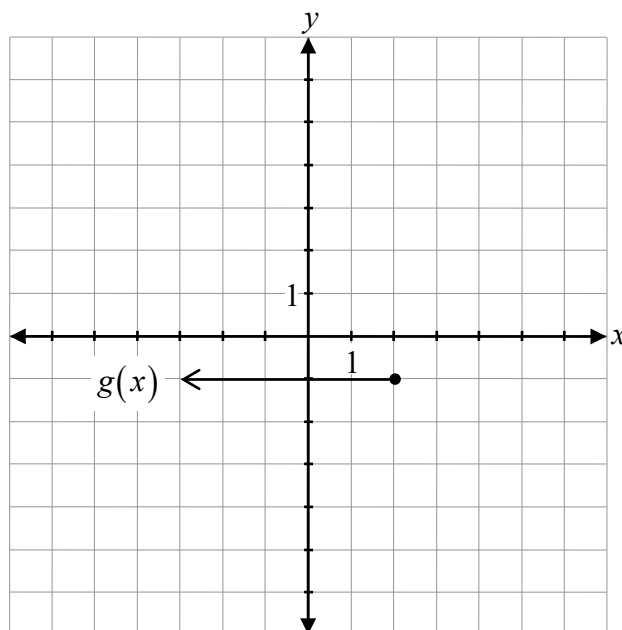
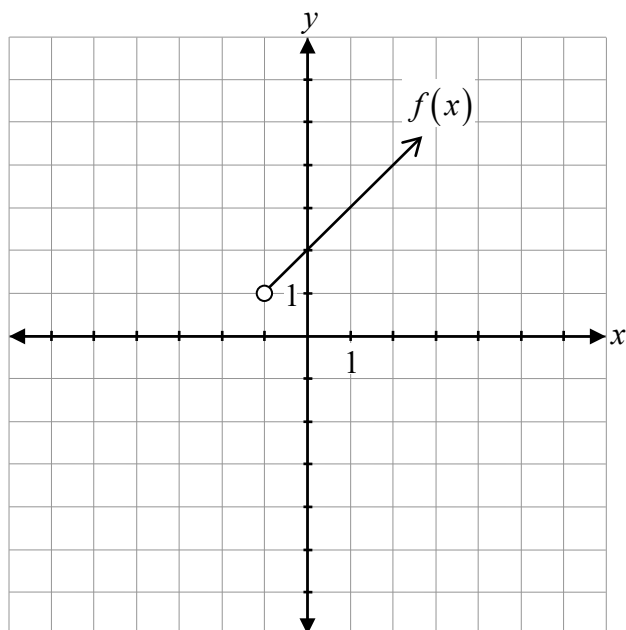


Question 16

1 mark

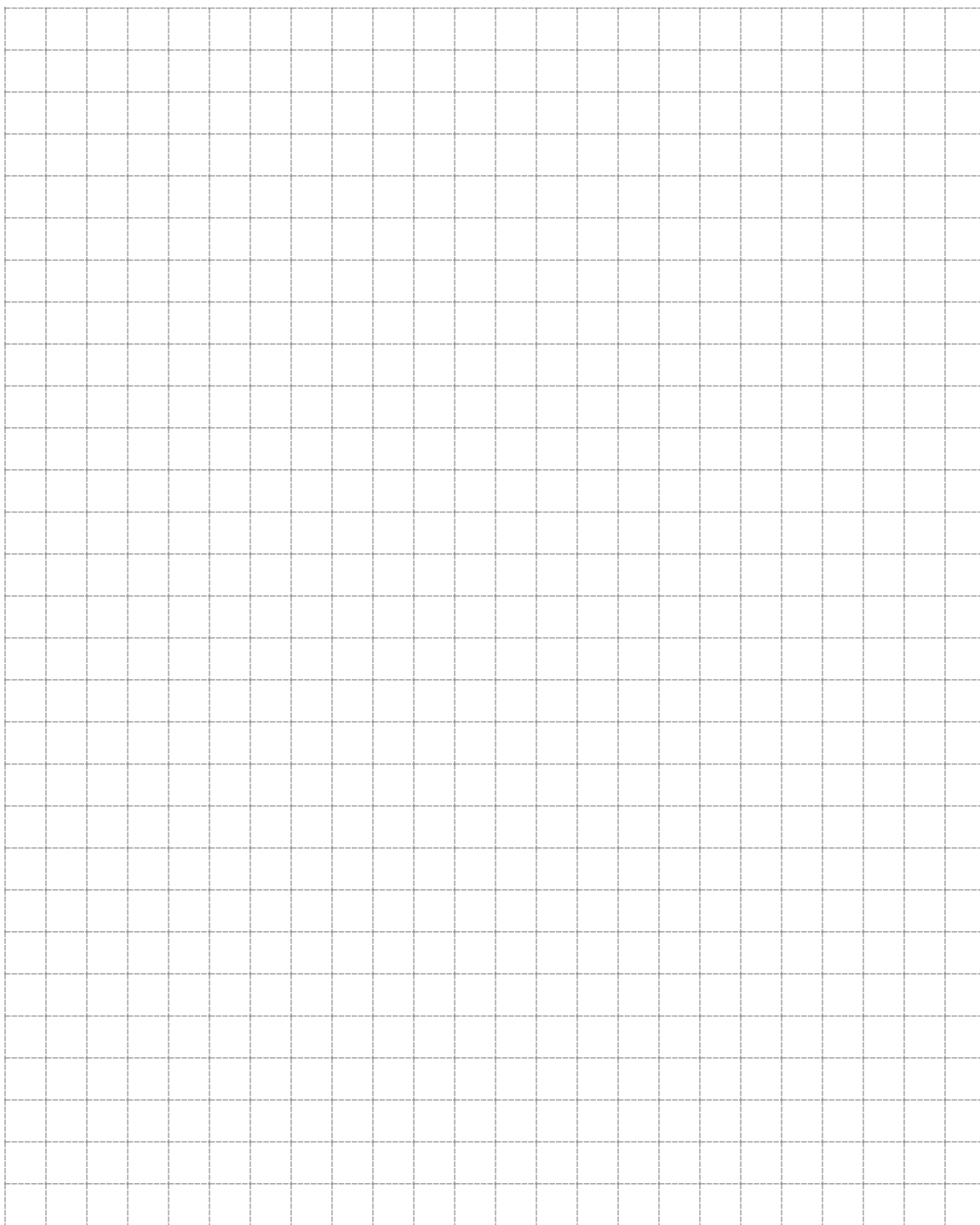
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Given the graphs of $f(x)$ and $g(x)$, state the domain of $(f + g)(x)$.



Domain: _____

No marks will be awarded for work done on this page.



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